

GrabU Platform

Hyperlocal On-Demand Grocery Delivery Ecosystem

A comprehensive product overview detailing system architecture, business logic, operational workflows, and core features for chartered accountants, investors, and operational partners.

Document Metadata

Platform Version:	v1.0 (Stable Release - Production Ready)
Ecosystem Apps:	Customer App (iOS/Android), Partner App (Android), Admin Panel (Web)
Core Technology Stack:	Flutter (Dart), Next.js (React/TypeScript), Supabase (PostgreSQL, Realtime, Auth)
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1. Executive Summary & Project Aim

The Challenge in Hyperlocal E-Commerce

Traditional e-commerce platforms operate on centralized warehousing systems with long delivery times. Hyperlocal e-commerce solves this by delivering goods from local shops to surrounding neighborhoods in minutes. However, retail grocery storefronts face severe bottlenecks: catalog creation is time-consuming, stock management is complex, and pricing fresh produce by weight creates checkout discrepancies. GrabU is engineered from the ground up to solve these retail challenges.

The Aim of GrabU

GrabU aims to empower local supermarkets and brick-and-mortar grocery stores to achieve digital transformation with zero setup friction. By utilizing a shared 'Master Product Catalogue' and simplified, automated configurations, shop owners can set up a fully stocked digital store within minutes.

Key business objectives include:

- Rapid Digital Onboarding: Eliminating manual data entry of barcodes, brands, and images.
- Zero-Overhead Server Scale: Utilizing Supabase Backend-as-a-Service to keep database costs at zero for initial operations.
- Real-Time Convenience: Delivering groceries within 15-30 minutes through an integrated partner-driver coordination flow.
- Operational Efficiency: Enabling accurate checkout configurations for weight-based produce (vegetables & fruits) with zero manual calculations required by store staff.

2. The Three-App Ecosystem

A Unified Operational Framework

The GrabU platform functions as a tightly coupled ecosystem comprised of three applications interacting in real-time through a centralized Supabase database. Each application targets a specific persona in the hyperlocal delivery workflow.

A. Customer Application (Mobile - Flutter/Dart)

The consumer-facing portal allows users to find nearby registered supermarkets, browse available inventory, place orders, and track deliveries. Key features include:

- Precise Geolocation Selection: Animated pin and map selector with full Nominatim search auto-suggestions.
- Dynamic Cart System: Seamlessly merges standard single-price items and weight-based variable items.
- Real-Time Order Tracking: Renders live progress from placement, shop packaging, to delivery agent coordinates.

B. Partner Application (Mobile - Flutter/Dart)

Designed for retail supermarket owners and their store managers to handle daily shop operations:

- One-Click Catalogue Seeding: Browse global presets categorized by standard supermarket groups and activate items instantly.
- Simplified Inventory Control: Manage retail selling prices, stock counts, and toggling of weight-based configurations.
- Order Dispatch Workflow: Push notifications for new orders, step-by-step picking list, and coordinate handoff to delivery partners.

C. Admin Control Panel (Web - Next.js/React/TypeScript)

The control tower for platform administrators to monitor health, manage data presets, and run analytics:

- Master Catalogue Management: Admin-only dashboard to insert, delete, or update global products, brands, and packaging details.
- Platform Order Drawer: Monitor and inspect details of all platform-wide sales transactions, including structured JSON delivery addresses and itemization.
- Partner Verification: Verify and activate new merchants on the platform.

3. Technical Architecture & Database Design

Backend-as-a-Service (BaaS) via Supabase

GrabU is designed to minimize server operation costs while providing real-time data sync. By offloading server infrastructure to Supabase, we obtain authentication, file hosting, and real-time PostgreSQL database synchronization out-of-the-box, operating entirely within free tier limits. This is highly attractive for startup operations and retail partners.

Core Database Schema

The database is built on relational integrity in PostgreSQL, optimized for retail inventory scaling:

- ``master_product``: The global presets table managed by platform administrators. Stores standard EAN barcodes, names, brands, packaging sizes, emojis, descriptions, categories, and official BigBasket CDN image URLs. RLS is configured specifically to permit public reads but restrict inserts to admin verification.
- ``shop``: Represents registered supermarkets, storing shop location (GPS coordinates), active status, and store operating hours.
- ``shop_product``: Stores the inventory listing for individual shops. Inherits core data from ``master_product`` via product presets, and maps shop-specific pricing (``selling_price``), availability (``stock_qty``), active status (``is_available``), and dynamic pricing tags (``weight_options``).
- ``orders`` & ``order_items``: Tracks checkout records. Delivery address details are serialized and stored as a structured JSON object to preserve exactly what the user selected on the map.

4. Key Operational Features & Business Logic

A. Hyperlocal Location Selector with Structured Address Parsing

The map-based geolocator solves coordinate-address discrepancies:

- Geolocator Fallback: Implements sequential, multi-tier GPS accuracy checks to prevent hardware timeout errors.
- Map Pin Animations: Provides visual lift-and-drop feedback when users pan the map viewport.
- Structured Storage: Address data is parsed and saved as a JSON block (Badge, House No., Area, City, Pin, Landmark, Phone) rather than a single string. This enables partners and administrators to render segmented, clean cards (e.g. Work or Home badge) and streamlines navigation for delivery drivers.

B. Preset Stocking Model (Master Catalogue)

Supermarket owners typically struggle with creating digital product files (taking packaging photos, writing descriptions). GrabU bypasses this entirely:

- Global Preset Database: Contains 167 real products (Staples, Dairy, Bakery, Snacks, Beverages, Cleaning, Personal Care, Instant Food, and Vegetables & Fruits).
- High-Fidelity Regional Brand Images: Populated with actual branded packaging image URLs (e.g. Eastern spices, Kitchen Treasures masalas, Double Horse puttu podi, Nirapara rice, Kera coconut oil, Milma dairy, Elite bakery) popular in Kerala, India.
- Instant Digital Inventory: Merchants filter presets by category, input their price/stock, toggle weight configuration, and tap save. The platform handles indexing instantly.

C. Dynamic Weight-Based Selling (Produce System)

Fruits and vegetables are sold by variable weight, which is historically hard to manage on standard e-commerce carts:

- Simplified Partner Switch: Instead of shop owners configuring multiple rows, they toggle a simple 'Sell by Weight' switch on any product card in the Partner App.
- Automated Customer Variants: The Customer App automatically generates 7 pre-set virtual weight variants: 100g (0.1x price), 250g (0.25x price), 500g (0.5x price), 750g (0.75x price), 1kg (1.0x price), 2kg (2x price), and 5kg (5x price).
- Zero-Code Checkouts: Tapping the item opens a modal listing all weight options with dynamic pricing. Added items map back to individual product IDs, processing cart inventory and accounting out-of-the-box.

5. Transaction Lifecycle & Business Workflow

Step-by-Step Transaction Flow

1. Shop Preset Stocking: A supermarket partner logs into the GrabU Partner App, browses the master catalogue (e.g., Staples, Dairy, Fresh Produce), inputs retail pricing, and enables products.
2. Customer Discovery: A customer launches the Customer App, sets their precise home location on the map, and selects the nearest supermarket. The inventory list loads automatically.
3. Cart Configuration & Checkout: The customer adds branded items and weight-based fresh produce (e.g. 500g Onion, 250g Ginger) to the cart, completes payment, and places the order. Address maps, names, and phone numbers are serialized to JSON.
4. Merchant Acceptance: The shop owner receives an order alert on the Partner App, opens the picking list showing quantities and packages the order. The system updates stock levels in real-time.
5. Courier Handoff: The packaged order is assigned to a nearby delivery driver who accesses the structured delivery coordinates and landmarks, navigating directly to the customer's doorstep.

Business Conclusion & Partner Alignment:

GrabU is an optimized, low-overhead e-commerce solution designed for immediate local deployment. By structuring addresses, offering pre-configured brand templates, and automating weight variant calculations, the platform reduces store operations labor by over 70% compared to typical manual catalog systems. This makes it highly scalable and ready for commercial retail integrations.